Interest or Rewards?

Children's reasoning about intrinsic and extrinsic motivation across task contexts

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Background

- Intrinsic and extrinsic motivation are central topics in developmental and educational research (Ryan & Deci, 2000).
- Intrinsic motivation is often linked to better learning, but performance effects also vary by task demands (Cerasoli et al., 2014; Howard,et al., 2021).
- The understanding of intrinsic and extrinsic motivation is linked to motivational outcomes and self-regulation (Hubley et al., 2024; Yu et al., 2023).

Research Question

- How do 7–10-year-olds evaluate intrinsic and extrinsic motivation, and predict their effects on learning outcomes?
- How does this reasoning vary across task demands (open-ended vs. closed-ended)?

Study 1

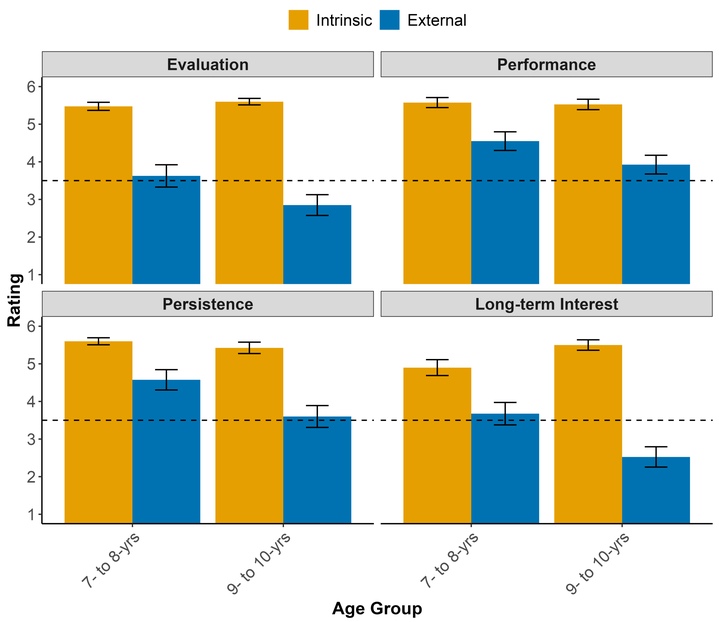
2 (motivation type: intrinsic, external) X
4 (evaluation/outcomes)

"I am going to learn because I love doing it — it's interesting and fun!"

"I am going to learn because my parents promised that they will buy my favorite gifts for me if I do.

Evaluation: Is it a good or bad reason to learn?
Performance: Will [name] become good or bad at doing it?
Persist: Will [name] keep trying on a hard task?
Lont-term interest: Will [name] keep learning after growing up?

Results



- N = 40 (7–8 years) and 40 (9–10 years)
- Intrinsic > external across age groups and measures ($ps < .001$, ds = 0.19 to 2.47)
- 9–10-year-olds evaluated external motivation more negatively than 7–8-year-olds ($ps < .017$; ds = -0.88 to -0.58)

Study 2

2 (motivation type: intrinsic, external) X
2 (Task type: open- vs. closed-ended)

CREATIVE & ORIGINAL

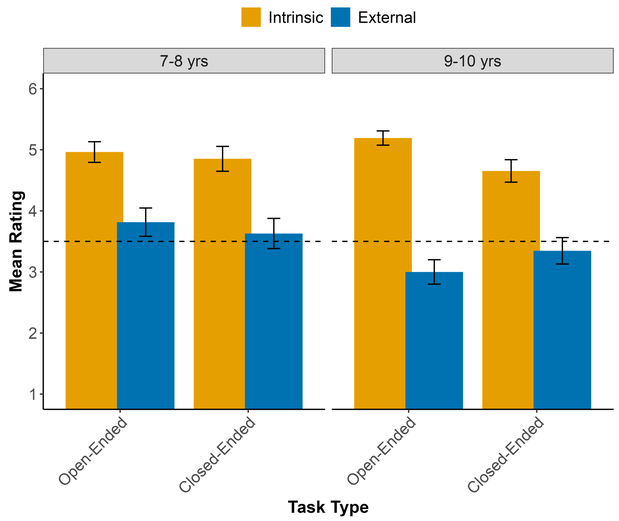
FAST & CORRECT

Riley thinks about the parts of the game that could be fun and interesting

Morgan thinks about the actual prizes Morgan could get from doing the game.

Task Performance:
How well would [name] do in the game?

Results



- N = 26 (7–8 years) and 27 (9–10 years)
- Intrinsic > extrinsic across task types ($ps < .001$, ds = 0.81 to 2.19)
- For intrinsic motivation, older children predicted greater performance benefits for open- than closed-ended tasks (p = .034, d = 0.37)

Summary

- Children aged 7–10 evaluated intrinsic motivation more positively and expected it to lead to better outcomes than external motivation.
- This pattern persisted across general learning contexts, open-ended, and closed-ended tasks.
- By ages 9–10, children recognized that intrinsic motivation may be less beneficial for performance on closed-ended than on open-ended tasks.

Reference

Cerasoli, C. P., Nicklin, J. M., & Ford, M. T. (2014). Intrinsic motivation and extrinsic incentives jointly predict performance: a 40-year meta-analysis. *Psychological bulletin*, *140*(4), 980–1008. https://doi.org/10.1037/a0035661
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